

Stress Detection using EEG signals: A review on current strategies and future aspects

1st Anmol Rattan Singh

Chitkara University Institute of Engineering and Technology
Chitkara University, Punjab, India
anmol.rattan@chitkara.edu.in

2nd Gurjinder Singh

Chitkara University Institute of Engineering and Technology
Chitkara University, Punjab, India
gurjinder.singh@chitkara.edu.in

3rd Nitin Saluja

Chitkara University Institute of Engineering and Technology
Chitkara University, Punjab, India
nitin.saluja@chitkara.edu.in

Abstract—Stress, a constant in modern life, harms mental and physical health. Electroencephalography (EEG) is useful for monitoring cerebral activity and identifying stress neural correlates because it is non-invasive. This review discusses EEG-based stress detection methodologies and future directions. The review begins with stress's physiological cause and EEG manifestation. It then discusses stress detection preprocessing, feature extraction, and machine learning algorithms. These methods include spectrum analysis, time-domain features, and deep learning models like CNN's, RNN's, and hybrid models. Stress detection systems benefit from data quality, artifact reduction, and feature selection. The paper also discusses EEG based stress detection in clinical diagnosis, occupational stress evaluation, and Brain-Machine Interface. It discusses real-world deployment issues and wearable EEG devices for continuous stress monitoring. Going ahead, the paper covers promising EEG stress detection technologies. This includes integrating multi modal data sources including physiological signals, eye-tracking, and behavioral data, creating tailored stress profiles, and using sophisticated machine learning and deep learning methods like transfer learning and AI.

Index Terms—EEG, stress detection, Deep learning (DL), Convolutional Neural Network (CNN), Feature Extraction; Machine Learning (ML)

I. INTRODUCTION

In contemporary society, a significant number of individuals are confronted with the imperative of overcoming various obstacles in order to successfully accomplish the objectives associated with their professional responsibilities within predetermined time frames. Consequently, there has been a significant increase in the prevalence of mental depression worldwide [1]. In many industrialised and developing countries, a significant proportion of the population is seeing a reduction in mental health problems [2]. The emergence of the COVID-19 pandemic has exacerbated the existing situation [3]. As a result, the ratio of mental patients to psychiatrists or psychologists has decreased even further. In

light of the ongoing crisis, scientists and researchers globally have dedicated their efforts to exploring alternative and automated methodologies for monitoring mental depression, with a particular focus on utilizing EEG data. Various machine learning and deep learning techniques, as well as diverse feature selection methods, are gaining prominence within this field. In a study conducted by Mumtaz and Qayyum, proposed that the use of 1D CNN and Long and Short-Term Memory (LSTM) models can achieve a high level of accuracy for the automated detection of unipolar depression. Specifically, the application of these models resulted in an accuracy of 98.32 percentage for the 1D CNN and 95.97 percentage for the LSTM model [4]. In their study, Cai et al. employed four fundamental techniques in deep and machine learning, namely K-Nearest Neighbors (KNN), SVM, Classification Trees (CT), and Artificial Neural Networks (ANN), utilizing a 10-fold cross-validation approach. The authors reported that KNN exhibited the highest performance across all datasets, achieving an accuracy rate of 79.27 percentage (Cai et al.). In their study, Seal et al. conducted two experiments, specifically the record-wise split and the subject-wise split, in order to investigate the tracking of depression using EEG data. They employed a convolutional neural network based on deep learning techniques for this purpose. A recognition rate of 99.37 percentage and 91.4 percentage was reported by the authors [6]. In their study, Zhu et al. employed the content-based ensemble method (CBEM) to analyze two distinct datasets, resulting in respective accuracies of 82.5 percentage and 92.65 percentage [7]. In their study, Xiao-Wei Wang et al. employed various feature selection techniques, including wavelet analysis, power spectrum analysis, and nonlinear dynamical analysis. The extracted features were subsequently utilized as input for the support vector machine (SVM) classifier. The researchers noted that Linear Discriminant Analysis (LDA) can yield an accuracy rate of up to 91.77 percentage [8]. In their study, Xiaowei Li et al. introduced a novel methodology for monitoring depression using electroencephalogram (EEG) data. The deep forest method was utilized to transform the

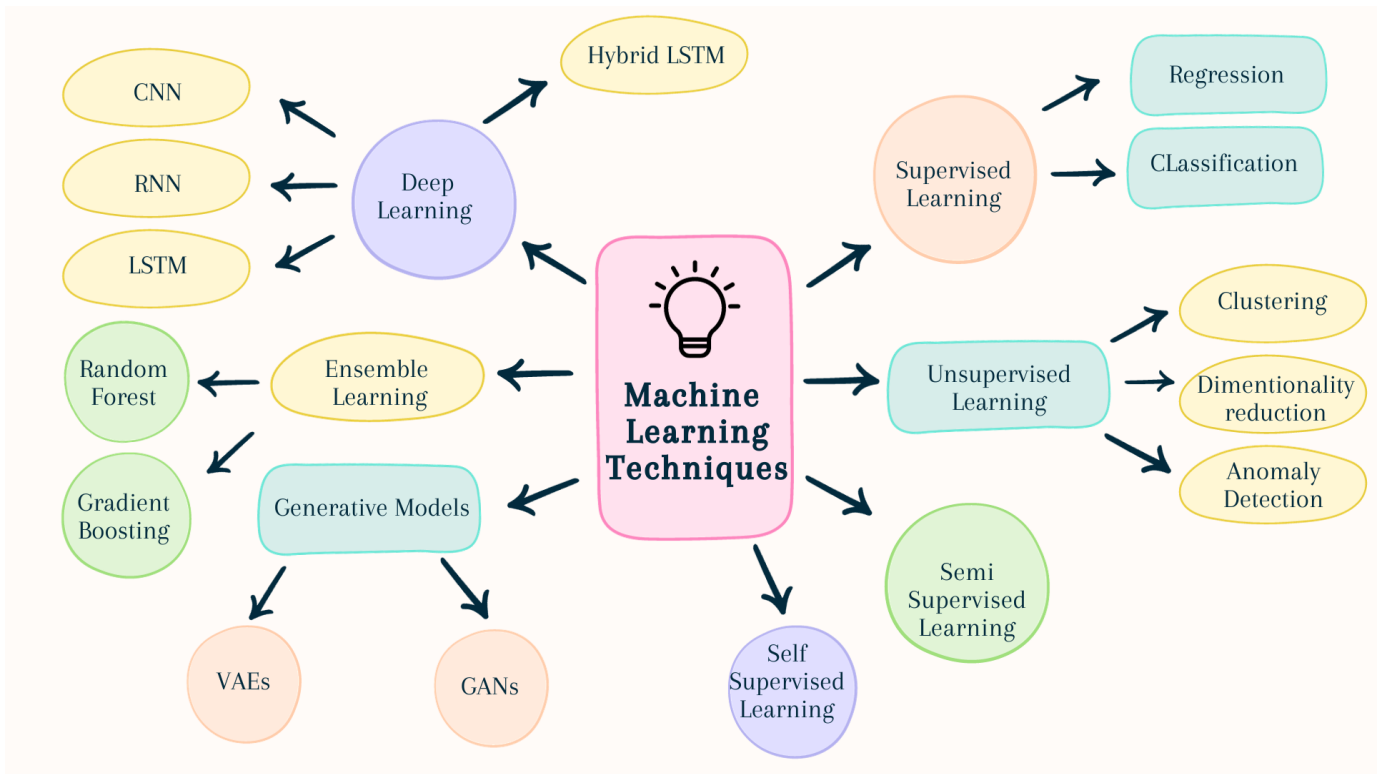


Fig. 1. Machine Learning Algorithms

original features into new features, which were subsequently employed as input for the SVM classifier. The researchers transformed the features into visual representations by incorporating specific EEG cap data into both the original and converted features. The CNN classifier has been successfully integrated with the input of data in the form of images. The highest achieved recognition rate on the ensemble model and power spectral density is 89.02 percentage. The Convolutional Neural Network (CNN) is a widely recognized deep learning model that has demonstrated an accuracy rate of 84.75 percentage according to a referenced study [9]. In their study, Natasha P. et al. presented two distinct approaches for feature extraction, specifically the Discrete Cosine Transformation and the Discrete Wave Transformation. The input features obtained were utilized to implement various classification algorithms, including Support Vector Machines (SVM), Linear Discriminant Analysis (LDA), k-Nearest Neighbors (KNN), Artificial Neural Networks (ANN), and Naïve Bayes classifier. This approach aimed to enhance the detection of mental stress from electroencephalogram (EEG) data [10]. Ran Bai et al. extracted linear features, non-linear features, and power spectral features in order to employ six distinct machine learning techniques, namely Support Vector Machines (SVM), K-Nearest Neighbors (KNN), decision trees, Naïve Bayes, random forests, and Logistic Regression. The results of their study demonstrated a greater level of recall compared to accuracy [11]. In their study, Ay et al. employed a composite model that combined Convolutional Neural Network

(CNN) and Long Short-Term Memory (LSTM) techniques to achieve enhanced accuracy in the detection of depression using Electroencephalogram (EEG) data [12]. According to the findings of Salama et al. (year), the utilization of a 3D CNN model has been identified as a viable approach for the purpose of emotion tracking using EEG data [13]. Sana Yasina et al. implemented an architecture based on Neural Networks to identify two distinct types of mental disorders, specifically Major Depressive Disorder and Bipolar Disorder, using EEG data. The authors developed a comprehensive research paper that presents a detailed examination of various EEG-based protocols, biomarkers, and publicly available datasets for the effective detection of mental disorders. Finally, a comprehensive explanation of techniques to enhance the dependability and efficiency of the suggested approaches has been provided [14]. This article presents a comparative study that aims to investigate the suitability of various machine and deep learning techniques, including MLP, CNN (referred to as CNN for brevity), RNN, RNN with LSTM, as well as supervised machine learning algorithms such as SVM and logistic regression, for tracking mental depression using EEG data. Additionally, the article emphasizes the importance of accelerating and finalizing the design and development of an expert system that can efficiently detect mental depression in a large population of patients within a short time span. The authors also advocate for more comparative studies in this domain, as they have societal benefits. Lastly, the authors want to make sure that all the current findings were discussed

TABLE I
COMPARATIVE ACCURACIES OF VARIOUS ML AND DL
MODELS

Models (Recognition)	Accuracy
MLP+SVM	96.42
CNN	56.4
RNN	96.4
RNN with LSTM	90.13
Fractal Dimention + SVM	67.06
OMTL + VonNeoman	71.14
Fractal Dimension + LSTM	67.06
FCDN	86.62
Deep Neural Network	80.5
Gradient-Boosted Decision Trees	81.3

and what future holds in this domain that could benefit the mankind from the silent killer, what other techniques that we can employ to make our findings more accurate, efficient and deliver faster results.

II. PRESENT ML AND DL TECHNIQUES

Various machine learning models find application in EEG signal analysis, with researchers exploring different approaches. For instance, in [2], an innovative method is proposed for categorizing emotional stress based on EEG signals, focusing on two primary regions in the valence-arousal space: calm-neutral and negatively excited. The LDA and SVM classifiers were employed, utilizing wavelet coefficients and chaotic invariants as input features. The classification accuracy for these two emotional stress states achieved 80.1% for LDA and 84.9% for SVM. Similar methods are described in [3], where the authors present a framework for measuring construction workers' stress levels using EEG headsets. This framework leverages an SVM algorithm to distinguish between low and high stress levels under various operational conditions. A different approach to wave analysis was proposed, one that makes use of the average values from Global Field Power for waves broken down into time units (consecutive minutes after the start of the event) rather than the more involved approach presented in [1]. These mean values proved effective for classifying emotions, specifically stress and relaxation, employing a MLP. Exceptional results were achieved, with classification accuracy exceeding 90%. When considering all brain waves, the highest accuracy recorded was 96.42%. Among specific wave categories, the Theta Gamma , and Alpha waves yielded the most favorable outcomes with values of (94.64 %), (80.36%), and (76.78%) respectively. Another noteworthy algorithm introduced by Hou et al. [6] employs the Stroop Color and Word Test (SCWT), capable of inducing four distinct levels of stress (relaxed, engrossed, stressful, and anxious). By incorporating statistical features and fractal dimension into the analysis, combined with SVM as the classifier, the method achieved a 67.06% accuracy rate. The SCWT was also utilized as a stressor in [7], where an automatic EEG-based stress recognition system was designed using an EMOTIV EPOC wireless device. This system successfully identified three stress levels with a 75% accuracy rate, relying on the relative difference

between beta and alpha power as a feature, and SVM as the classifier. A fully connected deep neural network (FC-DNN) and a convolutional deep learning neural network (deep CNN) were the two deep learning structures that the authors in [4],[13]. The maximum accuracy in identifying worker stress produced by the optimally configured DNN was 86.62%. When compared to their earlier manually created featurebased methods, it was at least 6% more accurate[5]. The authors in references [5] and [17] propose an EEG-driven stress recognition application that swiftly incorporates new input signals into its training dataset. To assess stress levels, the researchers employed diverse online multitask learning (OMTL) algorithms. EEG data was collected in two distinct settings: a controlled laboratory environment using wired EEG equipment and a real-world scenario utilizing a wearable EEG device. Among the various algorithms tested, the OMTL-VonNeuman algorithm demonstrated superior predictive accuracy, achieving 71.14% accuracy on the first dataset and 77.61% on the second. During examinations, EEG and electrocardiography (ECG) data were recorded from sixty participants. Cutting-edge metrics assessing linear and nonlinear dynamics of brain and cardiovascular signals were applied to extract relevant features. Employing an SVM classifier, the system effectively predicted participants' arousal perception with a 75% accuracy and valence perception with a 71.21% accuracy along the arousal and valence dimensions, respectively [8]. Multiple experiments were conducted using various EEG headsets to evaluate emotional responses induced within virtual reality (VR) environments, as documented in prior studies [10, 11, 12]. Notably, none of these studies delved into automatic emotion recognition. This research offers a unique contribution by bridging the realms of automatic response recognition and VR-based emotion induction: 1) Investigating the most discriminative brain wave parameters for automatic stress recognition induced within a virtual reality environment [17] is a primary focus. 2) Additionally, the study explores methods to reduce their number while preserving recognition quality, including a detailed analysis of classification performance for individual electrodes. A significant study detailed in [14] involved 19 healthy young individuals tasked with walking on a balance beam under virtual reality (VR) conditions designed to simulate high altitudes [18, 19]. The study recorded high-density EEG data, heart rate, electrodermal activity, the number of steps taken on the beam, and response times to auditory cues. Their findings suggest that virtual reality (VR) can provide users with physically realistic stress-inducing experiences. Fig. 1 shows the different Machine Learning Algorithms that we employ to analyse the EEG signals. Out of these techniques we have shortlisted few that have perform quite well as compared to others. So, the Table 1, will summarize the important results as per current research that we need to look upon and then we will can always improve on it to make our model faster if not accurate. As per table we have six most popular ML and DL techniques for which the results are shown. It is to be noted that about 30 % of the data is used in testing purposes while the bigger chunk of 70 % is for training the

TABLE II
DETAILED FINDINGS OF THE CURRENT TECHNIQUES

Study Reference	Objective	Methods	Key Findings	Accuracy	Model Used	Limitations	Future Directions
Kamińska et al. (2021)	Investigate the use of EEG signals for stress level classification in VR environments	Used alternating relaxing and stressful VR scenes with EEG monitoring on 28 healthy adults	High accuracy in stress level classification with CNNs; better results than conventional ML algorithms	96.42%	MLP and SVM classifiers	N/A	Suggests further exploration of EEG signal classification in VR for stress detection
Hou, X et al (2015)	Propose an algorithm for stress level recognition from EEG, using a Stroop colour-word test as a stressor	Experiment with 9 subjects to induce and record EEG data for 4 levels of stress	Different feature combinations and classifiers analyzed. Best results with fractal dimension, statistical features, and SVM	67.06% (4 levels), 75.22% (3 levels), 85.71% (2 levels)	SVM	Not mentioned	Could be applied for stress monitoring in high-responsibility professions like air traffic controllers
G. Jun et al (2016)	Design and implement an EEG-based stress recognition system using two effective stressors	Utilized the Stroop colour-word test and mental arithmetic test to induce low and high levels of stress, respectively.	Analyzed power band features and used beta-alpha power difference with SVM for high-accuracy stress recognition.	75% (3 levels), 88% (2 levels with Stroop test), 96% (2 levels with arithmetic test)	SVM	Not specified	Advocates broader use of EEG in stress detection and system advancements.
Jebelli, H. et al (2018)	Develop a framework to recognize construction workers' stress based on EEG activity using SVM	Utilized EEG data from workers performing tasks under different stress levels, with cortisol levels to label stress intensity	Achieved 71.1% accuracy in recognizing stress levels with Nonlinear SVM	71.1%	Nonlinear SVM	Real-world construction site conditions present challenges for EEG data acquisition	Potential application in real-time stress monitoring for construction workers to enhance workplace safety and health
Marín-Morales, J. et. al (2018)	Develop an emotion recognition system using Immersive Virtual Environments to evoke and recognize emotional states from neural and cardiac dynamics	Recorded EEG/ECG from 60 participants in virtual rooms to classify arousal-valence states using SVM.	Validated the use of Immersive Virtual Environments for eliciting and recognizing emotions; achieved notable accuracy in predicting arousal and valence	75.00% (arousal), 71.21% (valence)	SVM	Not specified	Opens novel applications in Architecture, Health, Education, and Videogames
Yang, H., Han, J., & Min, K. (2019)	Develop a multi-column CNN-based emotion recognition model from EEG signals	A band-pass filter was used and signals were down-sampled to 128 Hz. A weighted voting strategy collected individual recognition module decisions	The model addresses the non-stationary property of EEG signals and outperforms handcrafted feature- and deep neural network-based approaches in recognizing valence and arousal	81.4% (valence), 80.50% (arousal)	Multi-column CNN	The non-stationary nature of EEG signals still poses challenges	Further refinement of the model to handle the complexities of EEG signal analysis
Li, X., La, R., Wang, Y., Hu, B.,	Zhang, X. (2020)	Develop a novel approach for recognizing mild depression using EEG and graph theory	Used CNN on 2D EEG functional connectivity matrices and graph theory for network analysis to detect mild depression patterns.	Found increased path lengths and reduced network clustering in mild depression; CNN model highly accurate.	80.74%& CNN with functional connectivity matrices	-	Indicates CNN with connectivity matrices may help diagnose mild depression and support psychiatric research.

model to produce the results. Table 1 has the summarized results and scores of the tested ML and DL techniques. In next section we will discuss about changing our approach while improving the results in terms of less time taken or improving the overall accuracy in detecting the stress among humans. Table 2 explains in details about the literature reviewed and critical information that the ongoing and present technologies are offering in this domain of research.

III. FUTURE SCOPE

Since we know that EEG signals are continuous series and does not have anything to do with the image signals but it is shown that Hybrid models of DL have worked amazingly in terms of reducing time in generating the results in less time along with improved accuracy of the results as well. As it is known from above that a combo of MLP and SVM has produced the best results of 96.42 %. Therefore, it is certain that DL is best suited for extracting the best out of preprocessed data of EEG signals. Also it is confirmed that, no single model individually can generate better results. So we have to design a hybrid model comprises of multiple models in stages so that the results keep on getting improved with every stage. Therefore, a CNN model will work with an internal model like EfficientNet-LSTM-GRU. The above model has the capacity to work smoothly with image classification, but we have to modify the model so that it works to serve our purpose. A typical CNN-LSTM model will do the trick, or we can simply make some changes so that the code, especially in the CNN part, where the hidden layer can be modified as per our needs, Like Conv2D, layers can be configured. By default, 32 layers are there but we can simply increase or decrease the layers as per our needs and accuracy. In order to account for nonlinearity, the model employs an activation function called a Rectifier Linear Unit (ReLU). After the convolutional layer, a MaxPooling 2D layer was used to down sample the feature maps by decreasing the feature maps' spatial dimension while keeping the most important information. Lastly, the Flatten layer will be used to convert the 2D-generated feature maps to 1D vectors so that LSTM model can process it further to extract the vitals form the information. The subsequent LSTM layers are responsible for modeling the temporal dependencies in the EEG signals. This hybrid architecture combines both spatial (CNN) and temporal (LSTM) information, which can be effective for tasks like stress detection from EEG data. The LSTM layer is categorized as a recurrent neural network (RNN) layer, which has been specifically developed to effectively process sequential data. This characteristic makes it well-suited for tasks that involve time-series data, such as EEG signals. This LSTM model in this particular instance, has a total of 64 LSTM units. The aforementioned units are tasked with acquiring knowledge and discerning patterns inherent in the sequences of EEG signals. The LSTM layer is then configured to produce the complete sequence of outputs for every input sequence. In essence, it maintains the temporal information of the electroencephalogram (EEG) signals across the entire sequence. The LSTM layer is in charge of processing

the electroencephalogram (EEG) signal sequences that the preceding convolutional layers supply. The output of this layer is a sequence of hidden states, wherein each state encapsulates information pertaining to the input sequence up until that particular moment. Then the LSTM layer is able to transmit a sequence of hidden states to either the subsequent LSTM layer or the final layer. This facilitates the model's ability to comprehend and acquire intricate temporal patterns within the EEG data. In time-series data, such as EEG signals, the LSTM layer is crucial for capturing extended temporal dependencies and patterns. The incorporation of past information from EEG signals into the model facilitates the comprehension of the relationship between historical data and the present prediction. This is of utmost importance in stress detection, as EEG patterns are subject to temporal variations and may encompass significant insights into an individual's stress levels.

IV. CONCLUSION

In conclusion, EEG data analysis for mental depression detection has advanced and shown promise. Due to the COVID-19 pandemic and global depression, innovative automated monitoring methods are being researched. Researchers have used ML, DL, and feature selection methods to detect mental depression efficiently and accurately. Existing research has found several key findings. Numerous studies have found high accuracy rates using various methods. DL's potential in this domain was shown by one-dimensional CNN and LSTM models' 98.32% accuracy. Features selection methods like wavelet analysis and power spectrum analysis have also been effective, with LDA achieving 91.77% accuracy. Deep learning models like CNNs have achieved 84.75% accuracy. Moreover, stress detection from EEG data has been approached using MLP, CNN, RNN, RNN with LSTM, SVM, and LR algorithms. Each model has tested with varying degrees of accuracy, adding to the mental depression detection toolkit. Future prospects are bright for this field. CNN-LSTM hybrid models reduce processing time and improve accuracy. Further research should refine hybrid models to improve results. The study of VR-based stress induction and monitoring using EEG data offers promising prospects for improving mental depression detection systems. In conclusion, EEG-based mental depression detection research is improving our understanding of this important field. Researchers are using ML, DL, and innovative hybrid models to develop automated systems that can quickly and accurately identify mental depression, improving the lives of millions worldwide.

REFERENCES

- [1] Kamin'ska D, Smo'ika K, Zwolin'ski G. Detection of Mental Stress through EEG Signal in Virtual Reality Environment. *Electronics*. 2021; 10(22):2840. <https://doi.org/10.3390/electronics10222840>
- [2] Hosseini, S. A., Khalilzadeh, M. A., & Changiz, S. (2010). Emotional stress recognition system for affective computing based on bio-signals. *Journal of Biological Systems*, 18(spec01), 101-114.
- [3] Jebelli, H., Khalili, M. M., Hwang, S., & Lee, S. (2018, March). A supervised learning-based construction workers' stress recognition using a wearable electroencephalography (EEG) device. In *Construction Research Congress 2018* (pp. 40-50).

- [4] Jebelli, H., Khalili, M. M., & Lee, S. (2019). Mobile EEG-based workers' stress recognition by applying deep neural network. In *Advances in Informatics and Computing in Civil and Construction Engineering: Proceedings of the 35th CIB W78 2018 Conference: IT in Design, Construction, and Management* (pp. 173-180). Springer International Publishing.
- [5] H. Jebelli, M. Mahdi Khalili and S. Lee, "A Continuously Updated, Computationally Efficient Stress Recognition Framework Using Electroencephalogram (EEG) by Applying Online Multitask Learning Algorithms (OMTL)," in *IEEE Journal of Biomedical and Health Informatics*, vol. 23, no. 5, pp. 1928-1939, Sept. 2019, doi: 10.1109/JBHI.2018.2870963.
- [6] Hou, X., Liu, Y., Sourina, O., Tan, Y. R. E., Wang, L., & Mueller-Wittig, W. (2015, October). EEG based stress monitoring. In *2015 IEEE International Conference on Systems, Man, and Cybernetics* (pp. 3110-3115). IEEE.
- [7] G. Jun and K. G. Smitha, "EEG based stress level identification," *2016 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, Budapest, Hungary, 2016, pp. 003270-003274, doi: 10.1109/SMC.2016.7844738.
- [8] Mar'in-Morales, J., Higuera-Trujillo, J.L., Greco, A. *et al.* Affective computing in virtual reality: emotion recognition from brain and heartbeat dynamics using wearable sensors. *Sci Rep* 8, 13657 (2018). <https://doi.org/10.1038/s41598-018-32063-4>
- [9] Jensen, A. R. (1965). Scoring the Stroop test. *Acta psychologica*, 24(5), 398-408.
- [10] M. Horvat, M. Dobrinic', M. Novosel and P. Jerc'ic', "Assessing emotional responses induced in virtual reality using a consumer EEG headset: A preliminary report," *2018 41st International Convention on Information and Communication Technology, Electronics and Microelectronics (MIPRO)*, Opatija, Croatia, 2018, pp. 1006-1010, doi: 10.23919/MIPRO.2018.8400184.
- [11] Tarrant, J., Viczko, J., & Cope, H. (2018). Virtual Reality for Anxiety Reduction Demonstrated by Quantitative EEG: A Pilot Study. *Frontiers in psychology*, 9, 1280. <https://doi.org/10.3389/fpsyg.2018.01280>
- [12] J. Amores, R. Richer, N. Zhao, P. Maes and B. M. Eskofier, "Promoting relaxation using virtual reality, olfactory interfaces and wearable EEG," *2018 IEEE 15th International Conference on Wearable and Implantable Body Sensor Networks (BSN)*, Las Vegas, NV, USA, 2018, pp. 98-101, doi: 10.1109/BSN.2018.8329668.
- [13] Yang, H., Han, J., & Min, K. (2019). A Multi-Column CNN Model for Emotion Recognition from EEG Signals. *Sensors*, 19(21), 4736. <https://doi.org/10.3390/s19214736>
- [14] Peterson, S.M.; Furuichi, E.; Ferris, D.P. Effects of virtual reality high heights exposure during beam-walking on physiological stress and cognitive loading. *PLoS ONE* **2018**, *13*, e0200306
- [15] Katmah, R., Al-Shargie, F., Tariq, U., Babiloni, F., Al-Mughairbi, F., & Al-Nashash, H. (2021). A Review on Mental Stress Assessment Methods Using EEG Signals. *Sensors (Basel, Switzerland)*, 21(15), 5043. <https://doi.org/10.3390/s21155043>
- [16] Iriarte, J., Urrestarazu, E., Valencia, M., Alegre, M., Malanda, A., Viteri, C., & Artieda, J. (2003). Independent component analysis as a tool to eliminate artifacts in EEG: a quantitative study. *Journal of clinical neurophysiology*, 20(4), 249-257.
- [17] Can, Y. S., Arnrich, B., & Ersoy, C. (2019). *Stress Detection in Daily Life Scenarios Using Smart Phones and Wearable Sensors: A Survey*. *Journal of Biomedical Informatics*, 103139. doi:10.1016/j.jbi.2019.103139
- [18] Anjali, G. Singh and J. Kaur Sandhu, "Virtual and Augmented Reality Technology for the Treatment of Mental Health Disorders: An Overview," *2022 13th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, Kharagpur, India, 2022, pp. 1-5, doi: 10.1109/ICCCNT54827.2022.9984551.
- [19] Singh, G, Mantri, A, Sharma, O, Dutta, R, Kaur, R. Evaluating the Impact of Augmented Reality Learning Environment on Electronics Laboratory Skills of Engineering Students. *Comput Appl Eng Educ*. 2019; 27: 1361–1375. <https://doi.org/10.1002/cae.22156>
- [20] Li, X., La, R., Wang, Y., Hu, B., & Zhang, X. (2020). A Deep Learning Approach for Mild Depression Recognition Based on Functional Connectivity Using Electroencephalography. *Frontiers in Neuroscience*, 14. doi:10.3389/fnins.2020.00192
- [21] Wang, X.-W., Nie, D., & Lu, B.-L. (2014). Emotional state classification from EEG data using machine learning approach. *Neurocomputing*, 129, 94–106. <https://doi.org/10.1016/j.neucom.2013.06.046>
- [22] Yasin, S., Hussain, S. A., Aslan, S., Raza, I., Muzammel, M., & Othmani, A. (2021). EEG based Major Depressive disorder and Bipolar disorder detection using Neural Networks:A review. *Computer Methods and Programs in Biomedicine*, 202(106007), 106007. <https://doi.org/10.1016/j.cmpb.2021.106007>